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Performance Specification

V150-4.3 MW 50/60 Hz



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See general reservations, notes and disclaimers (including, Section 5, p.9) to this Performance Specification.

1 General Description

The Vestas V150-4.3 MW wind turbine is a pitch regulated upwind turbine with active yaw and a three-blade rotor. The Vestas V150-4.3 MW turbine has a rotor diameter of 150 m.

This Performance Specification contains power curves, Ct curves and sound curves for the following Power Optimized (PO) mode:

- PO2: V150-4.30 MW

2 Available Hub Heights

The standard turbine operated in 4.3 MW Power Optimized Mode PO2 is available with the hub heights listed below⁽²⁾:

Certification	Wind Class	Hub Height		
Tower type		Standard	Large diameter (split)	Large diameter (non-split) ⁽¹⁾
Design Evaluation Conformity Statement according to IEC61400-22	IEC S	105 m		155 m

Table 2-1: Type approval data and available hub heights

⁽¹⁾: These towers require special transport conditions as the bottom diameter is above 5 m and are not available as standard to the US/Canadian market but can be evaluated on a case-by-case basis.

⁽²⁾: Consult Vestas for specific tower height.

3 Operational Envelope and Performance Guidelines

Actual climate and site conditions have many variables and should be considered in evaluating actual turbine performance. The design and operating parameters set forth in this section do not constitute warranties, guarantees, or representations as to turbine performance at actual sites.

3.1 Climate and Site Conditions

The standard turbine operated in 4.3 MW Power Optimized Mode PO2 is designed for the wind climate conditions listed below. Values refer to hub height.

Wind Climate	IEC S
Extr Wind Speed (10 min average), V_{50}	37.5 m/s
Survival Wind Speed (3 s gust), V_{e50}	52.5 m/s

Table 3-1: Extreme design parameters – IEC

Wind Climate	IEC S
Wind Speed (10 min average), V_{ave}	6.5 m/s
Weibull Scale Factor, C	7.3 m/s
Weibull Shape Factor, k	2.0
Iref acc. to IEC 61400-1	0.14
Turbulence Intensity acc. to IEC 61400-1, Including Wind Farm Turbulence (@15 m/s) I90 (90% quantile)	15.7%
Wind Shear, α	0.20
Inflow Angle (vertical)	8°

Table 3-2: Average design parameters – IEC

3.1.1 Complex Terrain

Classification of complex terrain according to IEC 61400-1:2005 Chapter 11.2.

For sites classified as complex, appropriate measures are to be included in site assessment. Positioning of each turbine must be verified via Vestas Site Check.

3.1.2 Altitude

The turbine is designed for use at altitudes up to 1000 m above sea level as standard and optional up to 2000 m above sea level.

3.1.3 Wind Power Plant Layout

Turbine spacing is to be evaluated site-specifically. Spacing below two rotor diameters (2D) may require sector-wise curtailment.

NOTE As evaluation of climate and site conditions is complex, consult Vestas for every project. If conditions exceed the above parameters, Vestas must be consulted.

3.2 Operational Envelope – Wind

The turbines are offered with the High Wind Operation (HWO) control feature.

Values refer to hub height and are determined by the sensors and control system of the turbine.

Wind climate	IEC S
Hub height	105 m / 155 m
Cut-In, V_{in}	3 m/s
Cut-Out (10 min exponential avg.), V_{out}	24.5 m/s
Re-Cut In (10 min exponential avg.)	22.5 m/s

Table 3-3: Operational envelope – wind - IEC

3.3 Operational Envelope – Conditions for Power Curve and C_t Values (at Hub Height)

Consult Section 6, p. 10 for power curves and C_t values.

Conditions for Power Curve and C_t Values (at Hub Height)	
Wind Shear, α	0.00-0.30 (10 minute average)
Turbulence Intensity, I	6-12% (10 minute average)
Blades	Clean
Rain	No
Ice/Snow on Blades	No
Leading Edge	No damage
Terrain	IEC 61400-12-1
Inflow Angle (Vertical)	$0 \pm 2^\circ$
Grid Voltage	Nominal Voltage $\pm 2.5\%$
Grid Frequency	Nominal Frequency ± 0.5 Hz
Grid Active Power (at LV-side of turbine transformer)	As per tabulated values in Section 6.
Grid Reactive Power (at LV-side of turbine transformer)	Power Factor 1.0

Table 3-4: Conditions for power curve and C_t values

3.4 Sound Modes

The sound modes listed below are available for the turbine.

Sound modes			
Mode No.	Maximum Sound Level	Serrated trailing edges	Available hub heights
P02	104.9 dBA	Yes (standard)	105 / 155 m
P02-0S	107.6 dBA	No (option)	105 / 155 m

Table 3-5: Available sound performance

NOTE The turbine is as standard equipped with serrated trailing edges on the blades. Optionally, Mode P02-0S can be offered without serrated trailing edges mounted on the blades.

4 Drawings

4.1 Structural Design – Illustration of Outer Dimensions

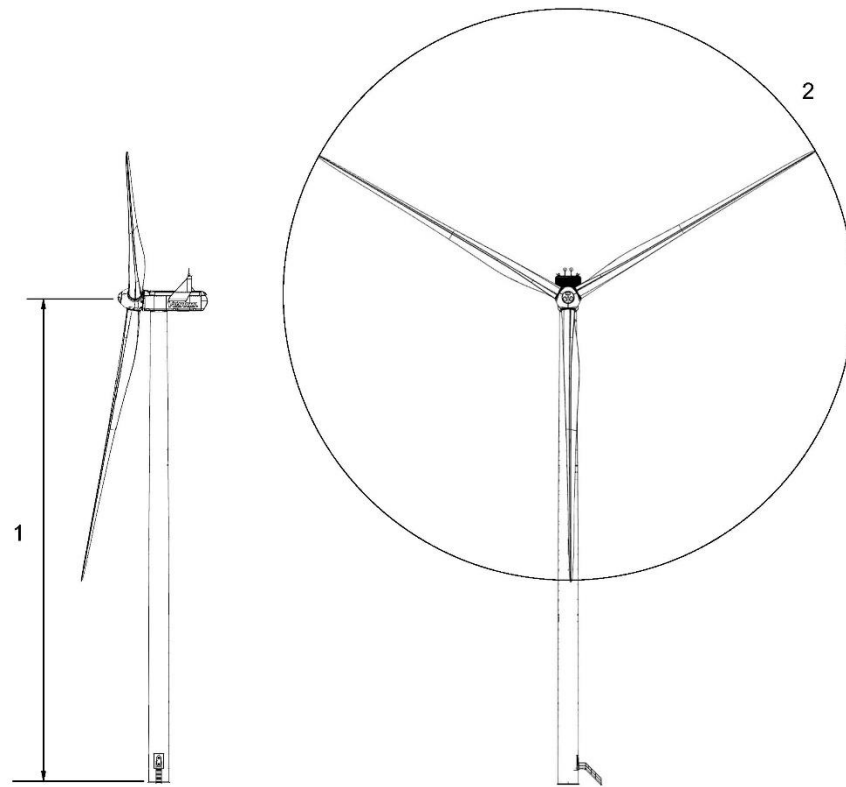


Figure 4-1: Illustration of outer dimensions – structure

1 Hub height 105/155 m

2 Diameter: 150 m

5 General Reservations, Notes and Disclaimers

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- The performance specifications described in this document apply to the current version of the V150-4.3 MW wind turbine. Updated versions of the V150-4.3 MW wind turbine, which may be manufactured in the future, may differ from these performance specifications. In the event that Vestas supplies an updated version of the V150-4.3 MW wind, Vestas will provide an updated performance specification applicable to the updated version.
- All listed start/stop parameters (e.g. wind speeds) are equipped with hysteresis control. This can, in certain borderline situations, result in turbine stops even though the ambient conditions are within the listed operation parameters.
- This document, Performance Specification, is not an offer for sale, and does not contain any guarantee, warranty and/or verification of the power curve and sound (including, without limitation, the power curve and sound verification method). Any guarantee, warranty and/or verification of the power curve and sound (including, without limitation, the power curve and sound verification method) must be agreed to separately in writing.

6 Power Curves, Ct Values and Sound Curves for Power Optimized (PO) Mode

6.1 Power Curves, Power Optimized Mode PO2/PO2-0S

Air density [kg/m^3]														
Wind speed [m/s]	1.225	0.95	0.975	1.0	1.025	1.05	1.075	1.1	1.125	1.15	1.175	1.2	1.25	1.275
3.0	81	51	54	56	59	62	65	67	70	73	76	78	84	86
3.5	172	122	127	131	136	140	145	149	154	158	163	167	176	181
4.0	285	210	217	223	230	237	244	251	258	264	271	278	291	298
4.5	424	317	327	337	346	356	366	375	385	395	404	414	433	443
5.0	596	451	464	477	491	504	517	530	543	557	570	583	609	622
5.5	808	615	632	650	668	685	703	720	738	755	773	790	825	843
6.0	1061	811	834	857	880	902	925	948	971	993	1016	1039	1084	1106
6.5	1360	1044	1072	1101	1130	1159	1188	1216	1245	1274	1303	1331	1389	1418
7.0	1711	1316	1353	1389	1425	1461	1496	1532	1568	1604	1640	1675	1746	1781
7.5	2106	1629	1673	1717	1761	1805	1848	1891	1935	1978	2021	2064	2149	2191
8.0	2549	1982	2034	2086	2138	2190	2242	2294	2345	2396	2447	2498	2599	2649
8.5	3021	2376	2437	2498	2559	2620	2678	2737	2795	2854	2909	2965	3074	3128
9.0	3470	2794	2860	2926	2992	3058	3119	3180	3241	3302	3358	3414	3523	3576
9.5	3839	3188	3254	3321	3388	3455	3514	3573	3632	3690	3740	3790	3881	3923
10.0	4099	3557	3620	3683	3746	3810	3857	3905	3952	4000	4033	4066	4124	4150
10.5	4228	3874	3922	3971	4019	4067	4096	4124	4153	4182	4197	4213	4238	4247
11.0	4287	4115	4144	4172	4200	4228	4240	4250	4262	4272	4277	4282	4289	4292
11.5	4298	4237	4247	4258	4269	4279	4283	4287	4291	4295	4296	4297	4299	4300
12.0	4300	4279	4283	4288	4292	4296	4297	4298	4299	4300	4300	4300	4300	4300
12.5	4300	4296	4297	4298	4299	4300	4300	4300	4300	4300	4300	4300	4300	4300
13.0	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
13.5	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
14.0	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
14.5	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
15.0	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
15.5	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
16.0	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
16.5	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
17.0	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
17.5	4296	4296	4296	4296	4296	4296	4296	4296	4296	4296	4296	4296	4296	4296
18.0	4230	4230	4230	4230	4230	4230	4230	4230	4230	4230	4230	4230	4230	4230
18.5	4102	4102	4102	4102	4102	4102	4102	4102	4102	4102	4102	4102	4102	4102
19.0	4003	4002	4002	4002	4002	4002	4002	4002	4002	4002	4003	4003	4003	4003
19.5	3895	3895	3895	3895	3895	3895	3895	3895	3895	3895	3895	3895	3895	3895
20.0	3770	3770	3770	3770	3770	3770	3770	3770	3770	3770	3770	3770	3770	3770
20.5	3607	3606	3606	3606	3606	3606	3606	3606	3607	3607	3607	3607	3607	3607
21.0	3408	3407	3407	3408	3408	3408	3408	3408	3408	3408	3408	3408	3408	3408
21.5	3180	3179	3179	3179	3179	3180	3180	3180	3180	3180	3180	3180	3180	3180
22.0	2917	2917	2917	2917	2917	2917	2917	2917	2917	2917	2917	2917	2917	2917
22.5	2645	2645	2645	2645	2645	2645	2645	2645	2645	2645	2645	2645	2645	2645
23.0	2363	2363	2363	2363	2363	2363	2363	2363	2363	2363	2363	2363	2363	2363
23.5	2070	2070	2070	2070	2070	2070	2070	2070	2070	2070	2070	2070	2070	2070
24.0	1782	1782	1782	1782	1782	1782	1782	1782	1782	1782	1782	1782	1782	1782
24.5	1561	1561	1561	1561	1561	1561	1561	1561	1561	1561	1561	1561	1561	1561

Table 6-1: Power curve, Mode PO2/PO2-0S (High Wind Operation)

6.2 Ct Values, Power Optimized Mode PO2/PO2-0S

Air density kg/m ³														
Wind speed [m/s]	1.225	0.950	0.975	1.0	1.025	1.05	1.075	1.1	1.125	1.15	1.175	1.2	1.25	1.275
3.0	0.886	0.890	0.890	0.890	0.889	0.889	0.888	0.888	0.888	0.887	0.887	0.886	0.886	0.885
3.5	0.845	0.851	0.850	0.850	0.849	0.848	0.848	0.847	0.847	0.846	0.846	0.846	0.845	0.845
4.0	0.828	0.827	0.827	0.827	0.828	0.828	0.828	0.828	0.828	0.828	0.828	0.828	0.828	0.828
4.5	0.825	0.826	0.826	0.826	0.826	0.826	0.826	0.826	0.825	0.825	0.825	0.825	0.824	0.824
5.0	0.820	0.824	0.823	0.823	0.823	0.822	0.822	0.822	0.821	0.821	0.821	0.820	0.820	0.819
5.5	0.817	0.821	0.820	0.820	0.820	0.819	0.819	0.819	0.818	0.818	0.818	0.817	0.817	0.816
6.0	0.813	0.818	0.817	0.817	0.816	0.816	0.816	0.815	0.815	0.814	0.814	0.813	0.812	0.812
6.5	0.810	0.815	0.815	0.814	0.814	0.813	0.813	0.812	0.812	0.811	0.811	0.810	0.809	0.808
7.0	0.806	0.814	0.813	0.813	0.812	0.811	0.811	0.810	0.809	0.809	0.808	0.807	0.806	0.805
7.5	0.801	0.811	0.810	0.809	0.808	0.808	0.807	0.806	0.805	0.804	0.803	0.802	0.800	0.799
8.0	0.792	0.806	0.805	0.804	0.803	0.801	0.800	0.799	0.798	0.796	0.795	0.794	0.790	0.789
8.5	0.764	0.802	0.800	0.797	0.795	0.792	0.789	0.785	0.781	0.778	0.773	0.769	0.759	0.754
9.0	0.702	0.773	0.767	0.761	0.755	0.749	0.742	0.736	0.729	0.723	0.716	0.709	0.696	0.689
9.5	0.628	0.712	0.704	0.697	0.690	0.682	0.675	0.667	0.660	0.652	0.644	0.636	0.620	0.611
10.0	0.550	0.650	0.641	0.633	0.625	0.616	0.607	0.598	0.589	0.579	0.570	0.560	0.540	0.530
10.5	0.472	0.588	0.577	0.567	0.557	0.547	0.536	0.525	0.514	0.503	0.493	0.482	0.462	0.451
11.0	0.403	0.525	0.513	0.501	0.489	0.478	0.466	0.455	0.444	0.432	0.423	0.413	0.394	0.385
11.5	0.345	0.458	0.446	0.434	0.422	0.410	0.400	0.390	0.380	0.370	0.362	0.354	0.338	0.330
12.0	0.299	0.396	0.386	0.375	0.365	0.354	0.346	0.337	0.329	0.320	0.313	0.306	0.293	0.286
12.5	0.261	0.344	0.335	0.326	0.317	0.308	0.301	0.294	0.286	0.279	0.273	0.267	0.256	0.251
13.0	0.231	0.301	0.294	0.286	0.278	0.270	0.264	0.258	0.252	0.246	0.241	0.236	0.226	0.221
13.5	0.205	0.266	0.259	0.253	0.246	0.239	0.234	0.229	0.224	0.218	0.214	0.209	0.201	0.197
14.0	0.183	0.237	0.231	0.225	0.219	0.213	0.209	0.204	0.199	0.195	0.191	0.187	0.180	0.176
14.5	0.165	0.212	0.207	0.201	0.196	0.191	0.187	0.183	0.179	0.175	0.171	0.168	0.162	0.158
15.0	0.149	0.190	0.186	0.181	0.177	0.172	0.169	0.165	0.161	0.158	0.155	0.152	0.146	0.143
15.5	0.135	0.172	0.168	0.164	0.160	0.156	0.153	0.149	0.146	0.143	0.140	0.138	0.132	0.130
16.0	0.123	0.156	0.152	0.149	0.145	0.142	0.139	0.136	0.133	0.130	0.128	0.125	0.121	0.118
16.5	0.112	0.142	0.139	0.136	0.133	0.129	0.127	0.124	0.121	0.119	0.117	0.114	0.110	0.108
17.0	0.103	0.130	0.127	0.124	0.121	0.118	0.116	0.114	0.111	0.109	0.107	0.105	0.101	0.099
17.5	0.095	0.120	0.117	0.115	0.112	0.109	0.107	0.105	0.103	0.101	0.099	0.097	0.094	0.092
18.0	0.087	0.109	0.107	0.104	0.102	0.099	0.097	0.095	0.094	0.092	0.090	0.088	0.085	0.084
18.5	0.078	0.098	0.096	0.093	0.091	0.089	0.088	0.086	0.084	0.082	0.081	0.080	0.077	0.076
19.0	0.070	0.088	0.086	0.084	0.082	0.080	0.079	0.077	0.076	0.074	0.073	0.072	0.069	0.068
19.5	0.064	0.080	0.078	0.076	0.074	0.073	0.071	0.070	0.069	0.067	0.066	0.065	0.063	0.062
20.0	0.058	0.072	0.070	0.069	0.067	0.066	0.065	0.063	0.062	0.061	0.060	0.059	0.057	0.056
20.5	0.052	0.064	0.063	0.062	0.060	0.059	0.058	0.057	0.056	0.055	0.054	0.053	0.051	0.050
21.0	0.046	0.057	0.056	0.055	0.053	0.052	0.051	0.050	0.049	0.049	0.048	0.047	0.045	0.045
21.5	0.041	0.050	0.049	0.048	0.047	0.046	0.045	0.044	0.044	0.043	0.042	0.041	0.040	0.039
22.0	0.036	0.044	0.043	0.042	0.041	0.040	0.039	0.039	0.038	0.037	0.037	0.036	0.035	0.034
22.5	0.031	0.038	0.037	0.036	0.035	0.035	0.034	0.033	0.033	0.032	0.032	0.031	0.030	0.030
23.0	0.027	0.032	0.032	0.031	0.030	0.030	0.029	0.029	0.028	0.028	0.027	0.027	0.026	0.026
23.5	0.023	0.027	0.027	0.026	0.026	0.025	0.025	0.024	0.024	0.024	0.023	0.023	0.022	0.022
24.0	0.019	0.023	0.022	0.022	0.022	0.021	0.021	0.021	0.020	0.020	0.020	0.019	0.019	0.019
24.5	0.017	0.020	0.019	0.019	0.019	0.018	0.018	0.018	0.018	0.017	0.017	0.017	0.016	0.016

Table 6-2: C_t values, Mode PO2/PO2-0S (High Wind Operation)

6.3 Sound Curves, Power Optimized Mode PO2/PO2-0S

Sound Power Level at Hub Height		
Conditions for Sound Power Level:	Measurement standard IEC 61400-11 ed. 3 Maximum turbulence at hub height: 30% Inflow angle (vertical): $0 \pm 2^\circ$ Air density: 1.225 kg/m^3	
Wind speed at hub height [m/s]	Sound Power Level at Hub Height [dBA] Mode PO2 (Blades with serrated trailing edge)	Sound Power Level at Hub Height [dBA] Mode PO2-0S (Blades without serrated trailing edge)
3	91.1	93
4	91.3	93.6
5	93.2	96.7
6	96.4	100.1
7	100.0	103.4
8	103.3	106.3
9	104.9	107.6
10	104.9	107.6
11	104.9	107.6
12	104.9	107.6
13	104.9	107.6
14	104.9	107.6
15	104.9	107.6
16	104.9	107.6
17	104.9	107.6
18	104.9	107.6
19	104.9	107.6
20	104.9	107.6
21	104.9	107.6
22	104.9	107.6
23	104.9	107.6
24	104.9	107.6

Table 6-3: Sound curves, Mode PO2/PO2-0S (High Wind Operation)